College Board AP® Chemistry

Your notes

VSEPR & Hybridization

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- VSEPR Theory
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- Sigma & Pi Bonds

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Your notes

VSEPR Theory

- The valence shell electron pair repulsion (VSEPR) theory is based on the principle that **electron groups** such as **single bonds, double bonds, triple bonds, or single electrons** repel one another through Coulombic forces
- The minimize this repulsion, the valence shell electrons should be placed as far apart in the space as possible
- VSEPR theory consist of three rules:

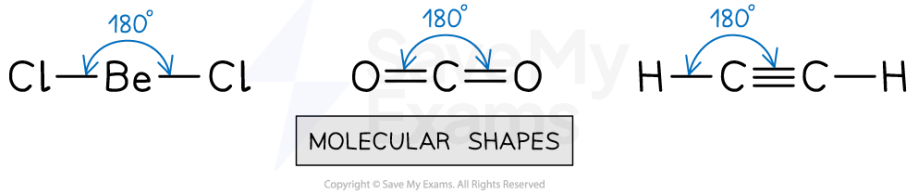
1. Bonding pairs and lone pairs should be arranged as far apart in space as possible
2. Double and triple bonds are **one** electron group
3. The Coulombic repulsion forces from lone pairs are stronger than the bonding pairs

- The **VSEPR theory** and the number of **electron groups in an atoms** is a useful tool that can be used to predict the structural and electron properties of the molecules

Two electron groups

- If there are two electron groups around the central atom, the angle between the bond is 180° which maximize their separation
- The geometry of this molecules is **LINEAR**
  - g. $BeCl_2$, CO_2, and ethyne (HC≡CH)

Examples of Linear Molecules

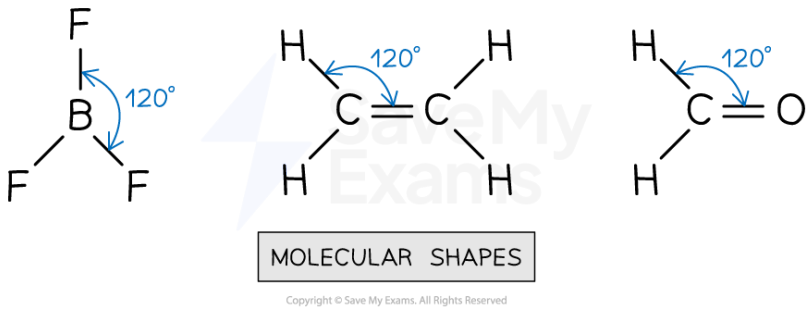


Beryllium chloride, carbon dioxide and ethyne all have two electron groups

Three electron groups

- If there are three electron groups around the central atom, the angle between the bonds is 120° which maximize their separation
- The geometry of this molecule is **TRIGONAL PLANAR**
  - E.g. BF_3, CH_2CH_2 and CH_2O

Examples of Trigonal Planar Molecules



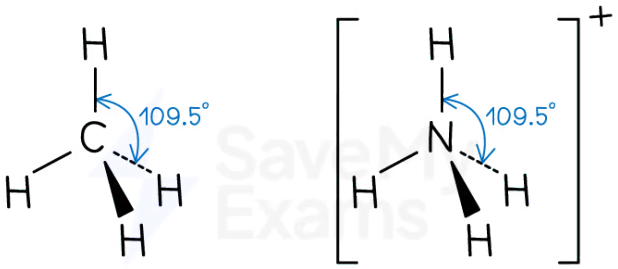
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Boron trifluoride, ethene and methanal all have three electron groups

Four electron groups

- Molecules with four electron groups have three dimensional geometries
- If there are four electron groups around the central atom, there are three case scenarios:
  - If the four electron groups are bonding groups, the angle between the bonds is 109.5° which maximize their separation
    - The geometry of this molecules is **TETRAHEDRAL**
    - E.g. CH_4, NH_4^+

Examples of Tetrahedral Molecules



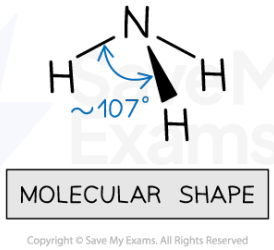
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Methane and ammonium ions have four electron domains

- If **three electron groups are bonding groups** and **one is a lone pair**, the angle is approximately 107° which is slightly less than 109.5° due to the Coulombic force of repulsion generated by the lone pair
  - The geometry of this molecules is **TRIGONAL PYRAMIDAL**
  - E.g. NH_3

NH_3 has a Trigonal Pyramidal Geometry

Your notes

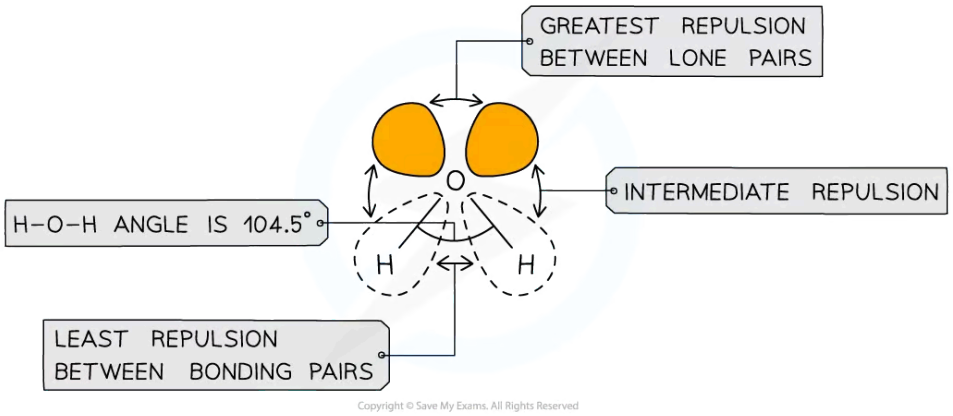


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The molecular geometry of ammonia

- If **two electron groups are bonding groups** and **two are lone pairs**, the angle is approximately 104.5° which is less than 109.5° due to the Coulombic force of repulsion generated by the two lone pairs

Bond Angles in Water

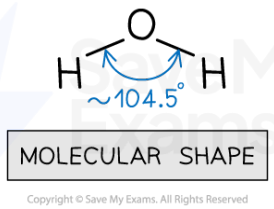


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The order of electron pair repulsion is lone pairs > lone pair: bonding pair > bonding pairs

- The geometry of this molecules is **BENT** or **ANGULAR**
- E.g. H_2O

Water Has a Bent Geometry



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The molecular geometry of water

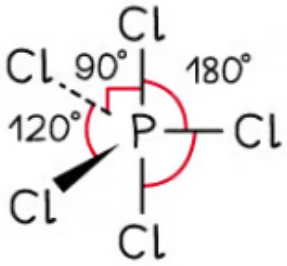
Five electron groups

Your notes

- Molecules with five electron groups have three dimensional geometries
- If there are five electron groups around the central atom, there are four case scenarios:
  - If the **five electron groups are bonding groups**, the angle between the equatorial bonds is 120° and the angle between the axial bonds is 90° which maximize their separation
    - The geometry of this molecules is **TRIGONAL BIPYRAMIDAL**
    - E.g. PCl_5

Your notes

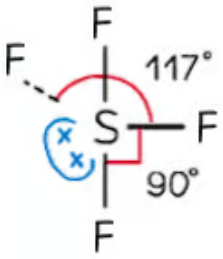
PCl_5 Has a Trigonal Bipyramidal Geometry



The molecular geometry of phosphorus pentachloride

- If **four electron groups are bonding groups** and **one of them is a lone pair**, the angle between the equatorial bonds is **slightly less than 120°** and the angle between the axial bonds is **slightly less than 90°** due to the Coulombic force of repulsion generated by the lone pair
  - The geometry of this molecules is **SEESAW**
  - E.g. SF_4

SF_4 Has a Seesaw Geometry



The molecular geometry sulfur tetrafluoride

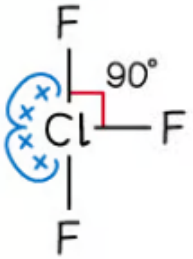
- If **three electron groups are bonding groups** and **two of them are lone pairs**, it is a flat molecule
- The angle between bonds is **slightly less than 90°** due to the Coulombic force of repulsion generated by the lone pairs
  - The geometry of this molecules is **T-SHAPED**

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- E.g. ClF_3


Your notes

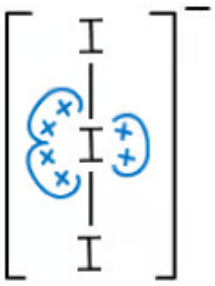
ClF_3 Has a T-shaped Geometry



The molecular geometry of chlorine trifluoride

- If **two electron groups are bonding groups** and **three of them are lone pairs**, it is a flat molecule
- The angle between bonds is **slightly 180°** due to the Coulombic force of repulsion generated by the lone pairs
  - The geometry of this molecules is **LINEAR**
  - E.g. I_3^-

I_3^- Has a Linear Geometry



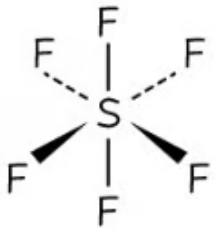
The molecular geometry of the triiodide ion

Six electron groups

- Molecules with five electron groups have three dimensional geometries
- If there are six electron groups around the central atom, there are three case scenarios:
  - If the **six electron groups are bonding groups**, the angle between the bonds is 90° which maximize their separation
    - The geometry of this molecules is **OCTAHEDRAL**
    - E.g. SF_6

SF_6 Has an Octahedral Geometry

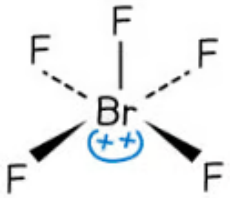
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The molecular geometry of sulfur hexafluoride

- If **five electron groups are bonding groups** and **one of them is a lone pair**, the angle between the bonds is **slightly less than** 90° due to the Coulombic force of repulsion generated by the lone pair
  - The geometry of this molecules is **SQUARE PYRAMIDAL**
  - E.g. BrF_5

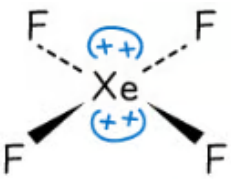
BrF_5 Has a Square Pyramidal Geometry



The molecular geometry of bromine pentafluoride

- If **four electron groups are bonding groups** and **two of them are lone pairs**, it is a flat molecule
- The angle between the bonds is 90° and the lone pairs are place at 180° minimizing the repulsion interaction between them
  - The geometry of this molecules is **SQUARE PLANAR**
  - E.g. XeF_4

XeF_4 Has a Square Planar Geometry



The molecular geometry of xenon tetrafluoride

In the table below it is a summary of the molecules that are part of the VSEPR

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Summary of the VSEPR theory

Electron groups	Bonding groups	Lone Pairs	Molecular Geometry	Bond angles
2	2	0	Linear	180°
3	3	0	Trigonal planar	120°
3	2	1	Bent	<120°
4	4	0	Tetrahedral	109.5°
4	3	1	Trigonal pyramidal	107°
4	2	2	Bent	104.5°
5	5	0	Trigonal bipyramidal	120° (equatorial) 90° (axial)
5	4	1	Seesaw	<120° (equatorial) <90° (axial)
5	3	2	T-shaped	<90°
5	2	3	Linear	180°
6	6	0	Octahedral	90°
6	5	1	Square pyramidal	<90°
6	4	2	Square planar	90°



Examiner Tips and Tricks

VSEPR theory is one of the most assessed topics in AP Chemistry. The molecular geometry, bond angles, bond order, relative bond energies, relative bond lengths, presence of dipole moment, and hybridization of valence orbitals can be predicted by using this model. Therefore, it is important that you understand its main principles



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and know how to explain them in terms of the **Coulombic repulsion between the electron groups**

Your notes

Molecular Geometry

Predicting molecular geometry

- The molecular geometry of any molecule can be determined following simple steps:

1. Draw an accurate Lewis structure for the molecule or ion
2. Count the number of electron groups
3. Count the number of bonding groups and the number of lone pairs
4. Apply the VSEPR theory to predict the molecular geometry

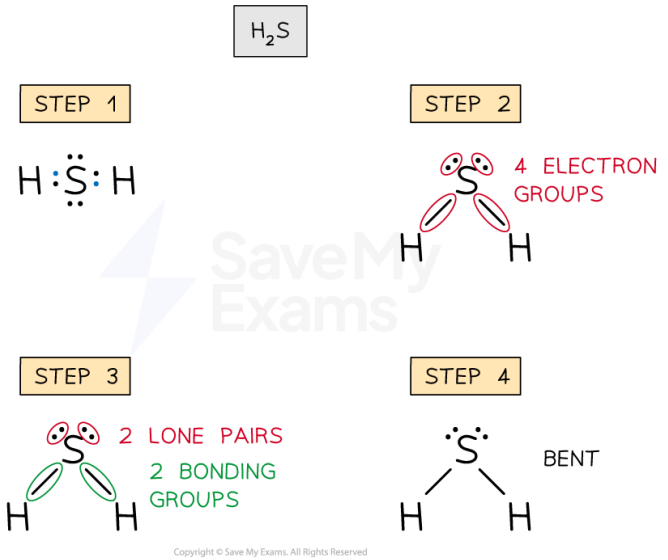
Worked Example

Predict the molecular geometry in the following molecules or ions:

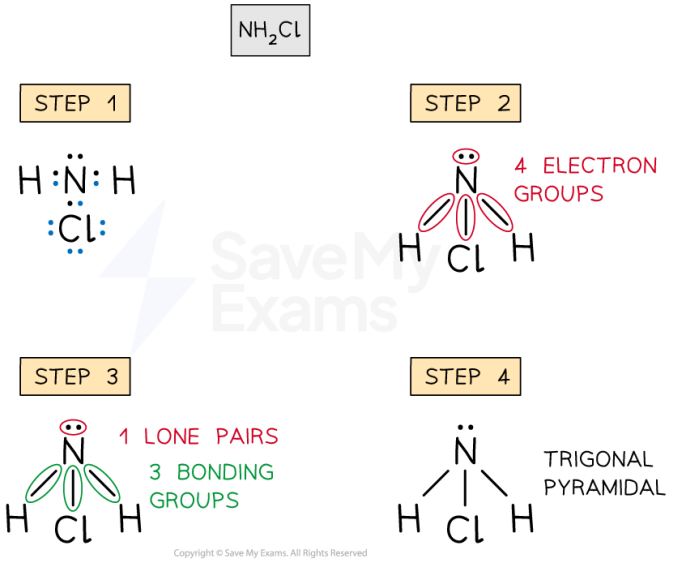
1. H_2S
2. NH_2Cl
3. ICl_4^-

Answers:

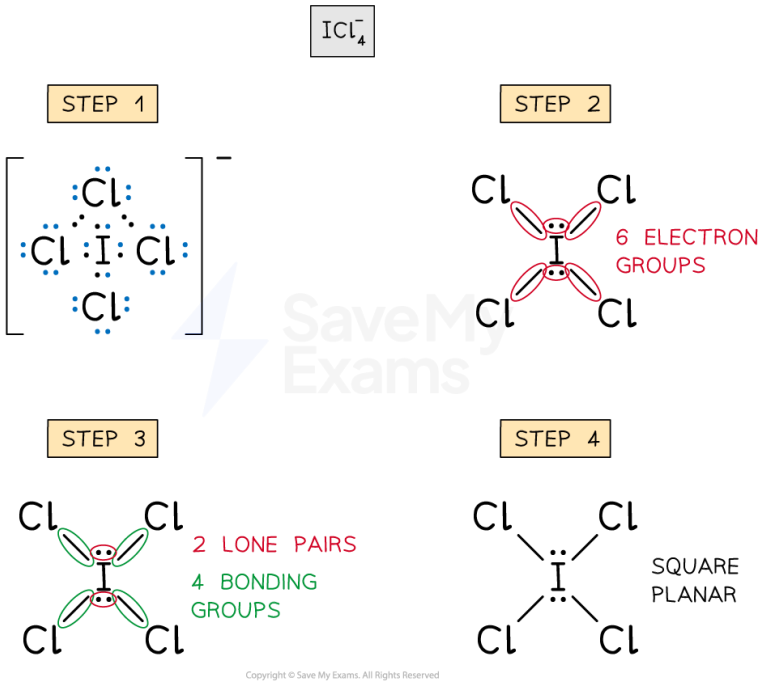
- **Answer 1:** The total number of valence electrons in H_2S is = 1 + 1 + 6 = 8, so there are four pairs of electrons around S



- **Answer 2:** The total number of valence electrons in NH_2Cl is = 5 + 1 + 1 + 7 = 14



- **Answer 3:** The total number of valence electrons in ICl_4^- is = 7 x 5 + 1 = 36



Bond Angles

Predicting bond angles

- The bond angle of any molecule can be determined following simple steps:

1. Draw an accurate Lewis structure for the molecule or ion
2. Count the number of electron groups and place them as far as possible
3. Count the number of bonding groups and the number of lone pairs
4. Apply the VSEPR theory to predict the bond angle

Your notes

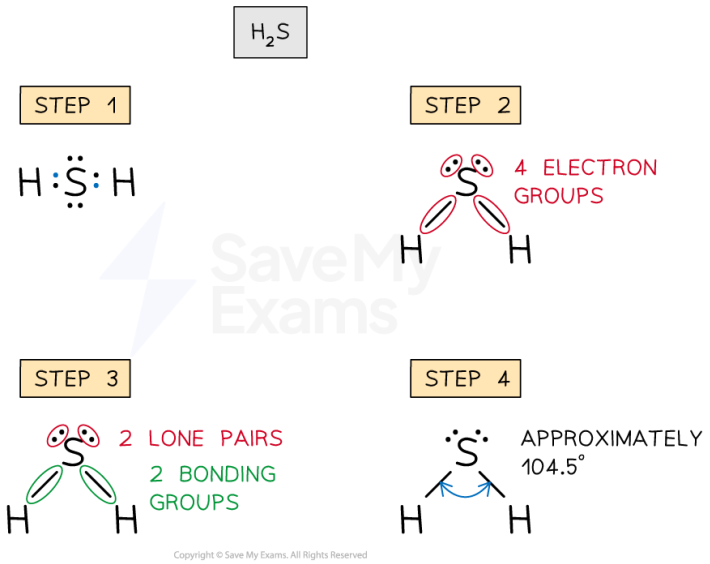
Worked Example

Predict the bond angles in the following molecules or ions:

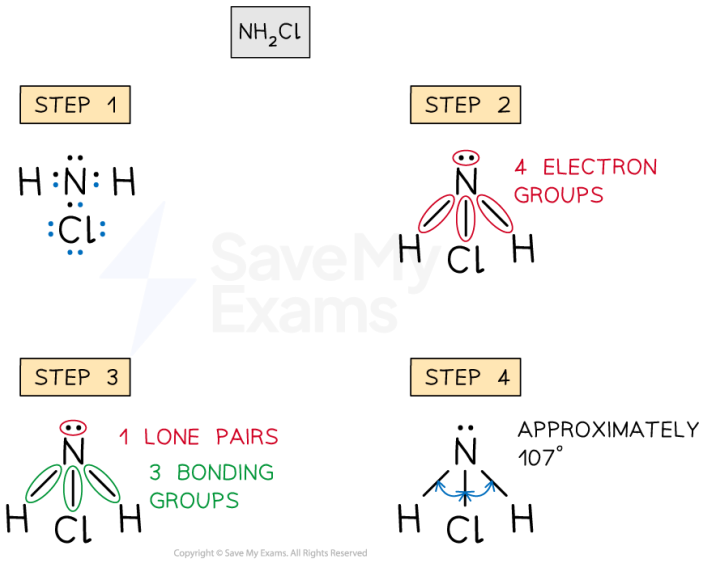
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Answers:

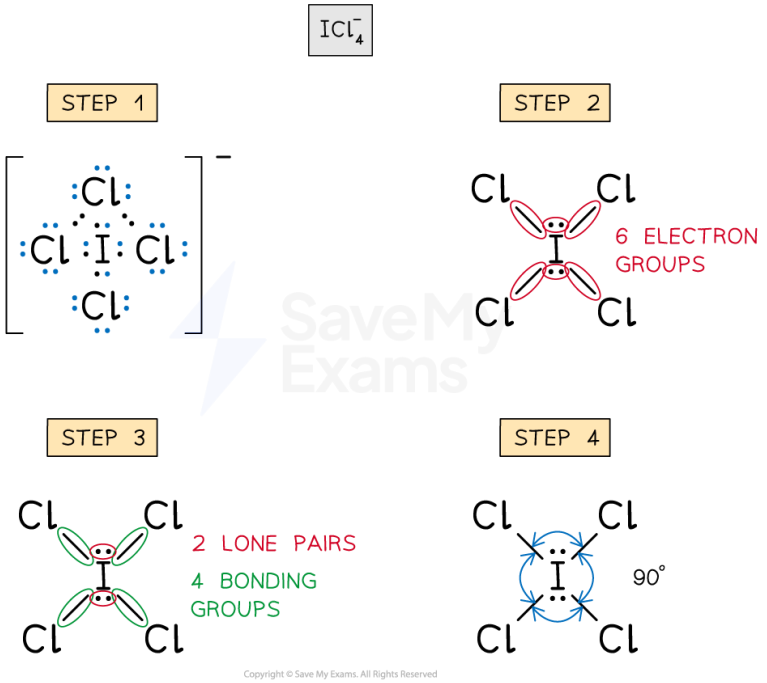
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- **Answer 3:** The total number of valence electrons in ICl_4^- is = 7 x 5 + 1 = 36



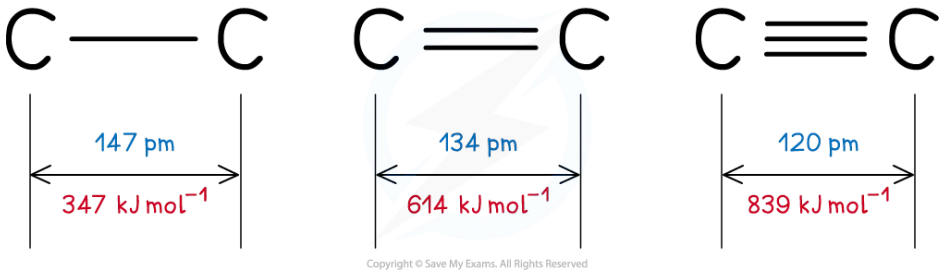
Your notes

Relative Bond Energies

Bond energy

- The **bond energy** is the amount of energy required to break 1 mol of bonds in the gas phase
  - It reflects the **strength of a chemical bond** between two atoms. The **higher the energy, the stronger the bond**
- It is influenced by factors such as bond type (ionic or covalent), bond length, and the presence of multiple bonds
- There are some important rules to follow when predicting the relative bond energy:
  - Single bonds are generally weaker than double bonds, and double bonds are weaker than triple bonds. Therefore, **single bonds have smaller bond energy**
  - The presence of multiple bonds increases bond strength because there is a greater Coulombic attraction between the electrons and the nuclei of the atoms

Bond lengths



Triple bonds has the highest bond energy (in red) and therefore they are the strongest ones

- **Bond order** is also a great chemical tool used to compare the bond energies between molecules

Your notes

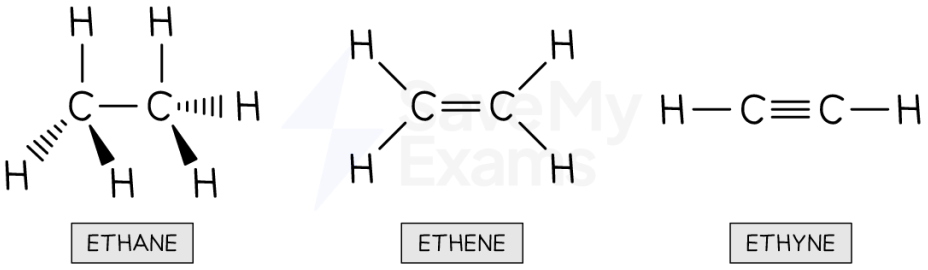
Bond order

- The **bond order** is the amount of bonding pairs of electrons between atoms
- **Higher bond orders** indicates **greater stability** and **higher bond energy**
- Bond order is usually an integer
- When a molecule has resonance, the bond order does not need to be an integer
- The bond order can be calculated by using three simple steps

1. Draw an accurate Lewis structures
2. Count the number of bonding pairs in the molecule
3. Count the number of bonding groups
4. Divide the number of bonding pairs by the number of bonding groups

Worked Example

Predict which of the following molecules has a the highest relative bond energy



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Answer:

In order to predict the molecule with the highest bond energy, the bond order of each molecule can be calculated

- **Ethane**
  - Bonding pairs = 7
  - Bonding groups = 7
  - Bond order = Bonding pairs / Bonding groups
  - Bond order = 7 / 7 = 1
- **Ethene**

- Bonding pairs = 6
- Bonding groups = 5
- Bond order = Bonding pairs / Bonding groups
- Bond order = 6 / 5 = 1.2

- **Ethyne**
  - Bonding pairs = 5
  - Bonding groups = 3
  - Bond order = Bonding pairs / Bonding groups
  - Bond order = 5 / 3 = 1.7

As seen in the calculations, the molecule with the **highest bond order** is **ethyne**. Since **higher bond order** correlates with **higher stability**, the molecule that needs the biggest amount of energy to break its bonds is ethyne.



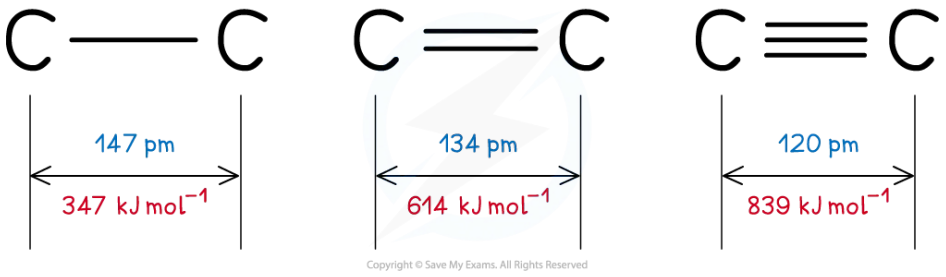
Your notes

Relative Bond Lengths

Bond length

- The **bond energy** refers to the **distance between the nuclei of two atoms that are chemical bond**ed
- It is influenced by factors such as bond type (ionic or covalent), bond length, the presence of multiple bonds, and the atomic radius of the atoms
- There are some important rules to follow when predicting the relative bond lengths:
  - If the **atomic radius** of the atoms involved **increases**, the **bond length will increase**
    - This occurs because the valence electrons are pulled with less Coulombic force of attraction by the nuclei of the atoms
  - The **presence of multiple bonds decreased the bond length** because there is a greater Coulombic attraction between the electrons and the nuclei of the atoms

Bond Lengths



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Triple bonds have the smallest bond length (in blue) because of the greater Coulombic forces of attraction

- **Bond order** is also a great chemical tool used to compare the bond lengths between molecules
  - If you want to know how to calculate the bond order, check the previous section about **Relative Bond Energy**

Your notes

Worked Example

Which of the following molecules will have the longest single covalent bond?

1. HCl
2. HF
3. HBr
4. HI

Answer:

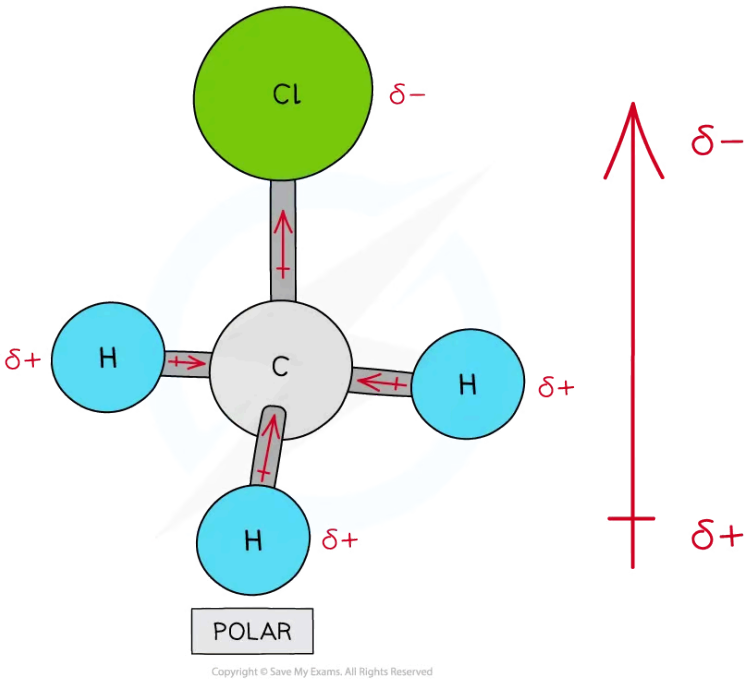
- The correct answer is **D**
  - This is because the iodine atom has the largest radius compared to Cl, F, and Br
    - This information can be obtained using the periodic table. Iodine is the down Group 17 under F, Cl and Br
  - Therefore, the coulombic attraction between the nucleus of the iodine atom and the valence electrons is the smallest, compared to the other halogens

Dipole Moments

- Dipole moment measures the separation of positive and negative charge within a molecule
- If a molecule exhibits dipole moments, it is said that the molecule is polar
- There are two important factors to consider if a molecule is polar or not:
  - The presence of polar covalent bonds in the molecule
  - The molecular geometry
- There are cases of molecules that have polar covalent bond, but their polarity its neutralized by the molecular geometry
  - This occurs because the atoms are arranged in a particular way, so the individual dipole moments cancel each other out
  - You can think of a dipole moment as a "tug of war game". If the atoms pull the electrons with the same force in opposite directions, the net dipole moment of the molecule is zero
  - E.g. The molecule below CH_3Cl is a polar molecule. The net dipole moment of the molecule points towards the chlorine atom which attracts the electrons with the strongest Coulombic force of attraction

The CH_3Cl Molecule

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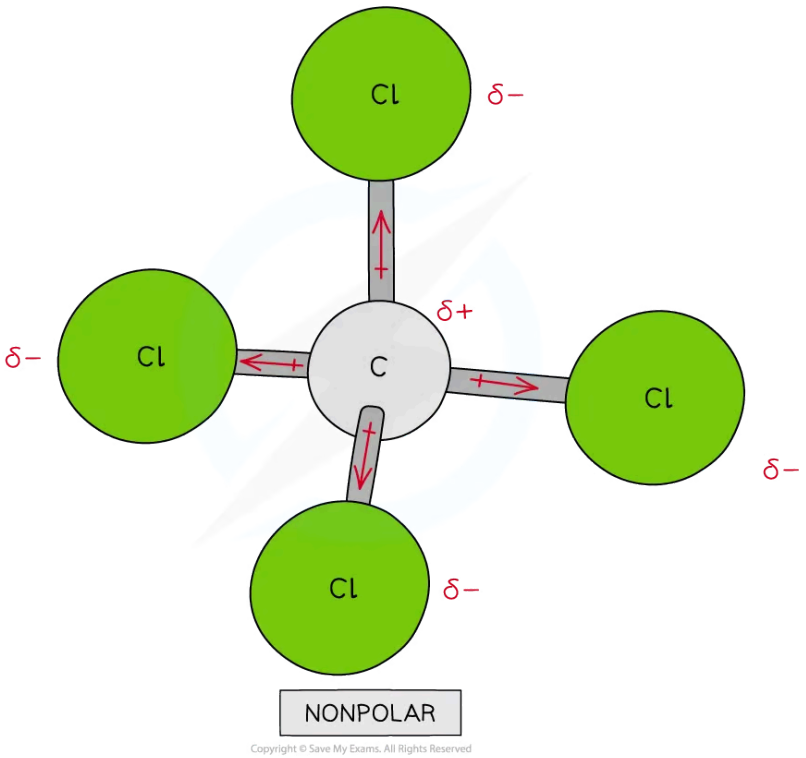
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Your notes

There are four polar covalent bonds in CH_3Cl which do not cancel each other out causing CH_3Cl to be a polar molecule; the overall dipole is towards the electronegative chlorine atom

- E.g. The molecule below CCl_4 is a **nonpolar** molecule. Even if the C-Cl bonds are highly polar due to the difference in electronegativity between C and Cl, the **net dipole moment cancels out** because all the chlorine nuclei are pulling the electrons with the same Coulombic force of attraction

The CCl_4 molecule



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Though CCl_4 has four polar covalent bonds, the individual dipole moments cancel each other out causing CCl_4 to be a nonpolar molecule

Your notes

Your notes

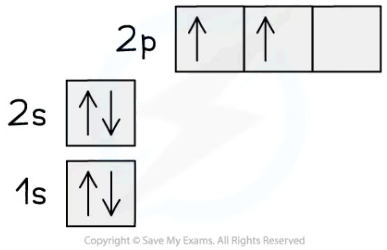
Hybridization of Valence Orbitals

- Covalent bonds are formed when atomic orbitals overlap that together contain a pair of electrons
- Hybridization occurs when atomic orbitals (s and p) are combined to form hybrid orbitals
- Hybrid orbitals are still localized in the individual atoms, but their shapes and energies are different
- The hybridization of valence orbitals is important to explain the molecular geometry of different carbon compounds

sp^3 hybridization

- Carbon ground state electron configuration is defined by $1s^2 2s^2 2p^2$
- It is represented by the following box diagram

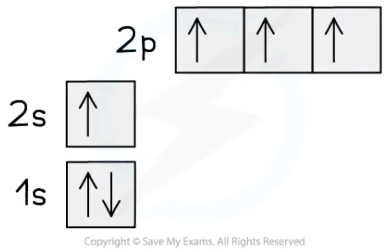
Box Diagram in Ground State



Orbital box diagram for carbon in the ground state

- By looking at the box diagram, there are two unpaired electrons which implies that only two covalent bonds can be formed
- The 2s electrons are paired so there is no reason that suggest that they will be involved in bonding
- The study of carbon compounds shows that the carbon atom tend to form **four covalent bonds**
- Since, 2s and 2p subshells are close in energy, one electron of the 2s subshell can be promoted to the 2p subshell giving the following new arrangement

Box Diagram in Excited State

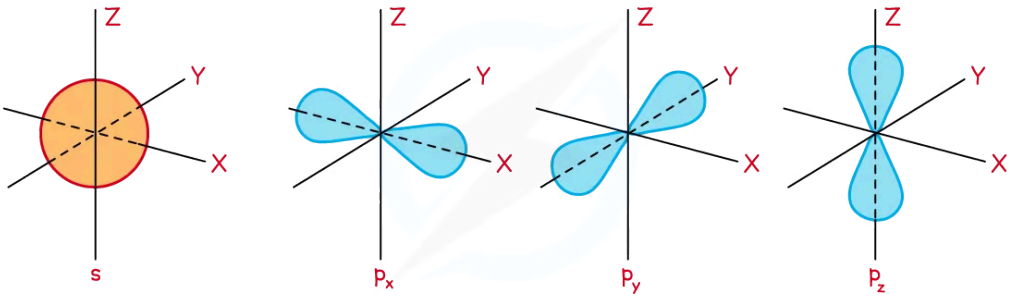


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Orbital box diagram for carbon in when a 2s electron is promoted to the 2p subshell

- Therefore, the s orbital from the 2s subshell and the three orbitals from the 2p subshell will merge together to form four hybrid orbitals
  - It is important to remember the shapes of the orbital involved in the hybridization
  - An **s orbital** has an **spherical shape**
  - A **p orbital** is **dumbbell or figure-of-eight shaped**
    - There are three p orbitals depending on their orientation in the xyz plane
    - They are perpendicular between themselves

s and p Orbitals



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The s orbital is spherical and the three dumbbell-shaped p orbitals lie at right angles to each other

- This new hybrid orbitals will be of **equal energy** and this explain how four covalent bonds are formed
- They are named **sp^3** hybrid orbitals, because it is the combination of **one s orbital** and **three p orbitals**

sp^3 hybrid orbitals



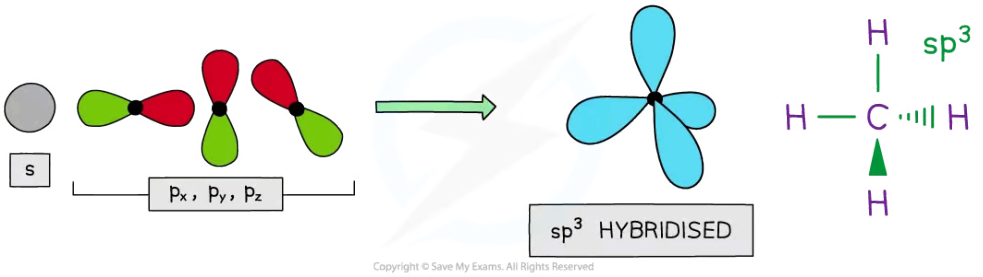
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Your notes

Orbital box diagram for carbon showing sp3 hybrid orbitals

- These new hybrid orbitals have ¼ s character and ¾ p character. Therefore, their shape would have a club shape, like an enlarged p orbital
- The **four sp^3 hybrid orbitals** can be arranged in space within a tetrahedron. Therefore, this is the origin of the 109.5° angle of the molecular geometry
- The carbon atom in methane shows **sp^3**
- The **sp^3 orbitals** in the carbon atom, overlaps with the s orbital of hydrogen forming four equal covalent bonds

Formation of sp^3 Hybridized Orbitals



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4 x sp^3 hybrid orbitals are formed from one s orbital and three p orbitals

- It is not just bonding pairs of electrons that can be accommodate in the hybrid orbitals, lone pairs can also be present
  - One of the clearest examples is the ammonia molecule. The nitrogen atom is **sp^3 hybridized,** but **three of the hybrid orbitals are used to arrange bonding pairs** and **one of them for a lone pair**

sp^2 hybridization

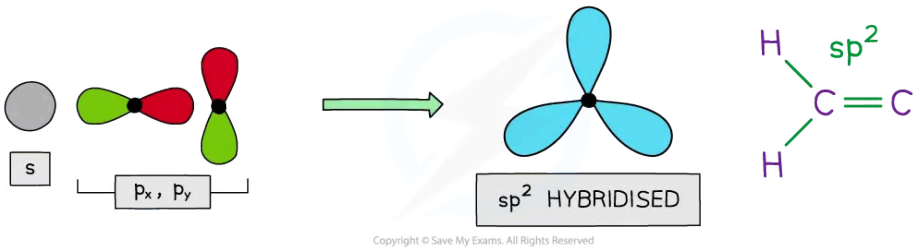
- **sp^2** hybrid orbitals are formed when one s orbital and two p orbitals are combined together
- One of the p orbitals of the 2p subshell remains unchanged after **hybridization**
- The **sp^2 orbitals have the same energy,** while the p orbital from the 2p subshell is higher in energy than the hybridized orbitals
- These new hybrid orbitals have ⅓ s character and ⅔ p character
- The **three sp^2 hybrid orbitals** can be arranged in space within a trigonal planar geometry. Therefore, this is the origin of the 120° angle of the molecular geometry
- The carbon atom in alkenes shows **sp^2 hybridization** when it forms two single bonds and a double bond with another carbon
- The **sp^2 orbitals** in the carbon atom overlap with the s orbital of two hydrogen atoms and one hybrid **sp^2 orbital of the adjacent carbon** forming three equal sigma bonds

Your notes

- The double bond is created when the unhybridized p-orbitals from adjacent carbons sideways overlap
    - Sideways overlap is a lateral interaction between the p-orbitals of two adjacent atoms

Your notes

Formation of sp^2 Hybridized Orbitals



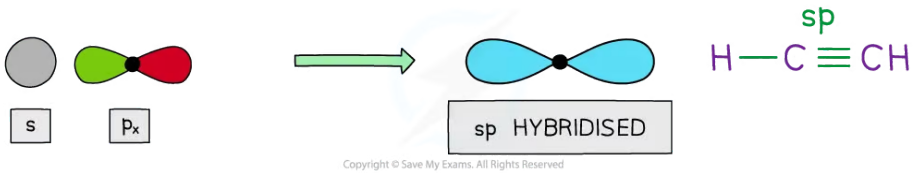
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3 x sp^2 hybrid orbitals are formed from one s orbital and three p orbitals

sp hybridization

- **sp** hybrid orbitals are formed when one s orbital and one p orbital are combined together
- Two of the p orbitals of the 2p subshell remains unchanged after **hybridization**
- The **sp orbitals have the same energy,** while the two p orbitals from the 2p subshell is higher in energy than the hybridized orbitals
- These new hybrid orbitals have ½ s character and ½ p character
- The **two sp hybrid orbitals** can be arranged in space forming a linear geometry. Therefore, this is the origin of the 180° angle of the molecular geometry
- The carbon atom in alkynes shows **sp hybridization** when it forms one single bond and a triple bond with another carbon
- The **sp orbitals** in the carbon atom overlap with the s orbital of the hydrogen atom and one hybrid **sp^2 orbital of the adjacent carbon** forming two equal sigma bonds
- The triple bond is created when the two unhybridized p-orbitals from adjacent carbons sideways overlap
    - Sideways overlap is a lateral interaction between two adjacent atoms

Formation of sp Hybridized Orbitals



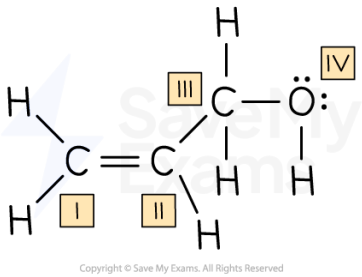
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2 x sp hybrid orbitals are formed from one s orbital and two p orbitals

Your notes

Worked Example

Compound X has the following Lewis diagram:



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Lewis diagram of compound X

What is the hybridization of the atoms I, II, III, and IV?

Answer:

- **Atom I**
  - The carbon atom is forming two single bonds and a double bond. The presence of the double bond indicates that it has an **sp^2 hybridization**
- **Atom II**
  - The carbon atom is forming two single bonds and a double bond. The presence of the double bond indicates that it has an **sp^2** Additionally, it is and adjacent carbon atom to an sp^2 carbon
- **Atom III**
  - The carbon atom is forming four single bonds.Therefore, it has an **sp^3 hybridization**
- **Atom IV**
  - The oxygen atom is forming **two single bonds** and it has **two lone pairs**. Even if the the atom is just forming two bonds, the oxygen atom has an **sp^3 hybridization** because two of the hybrid orbitals are used for bonding electrons and the other two are used for nonbonding electrons

Your notes

Sigma & Pi Bonds

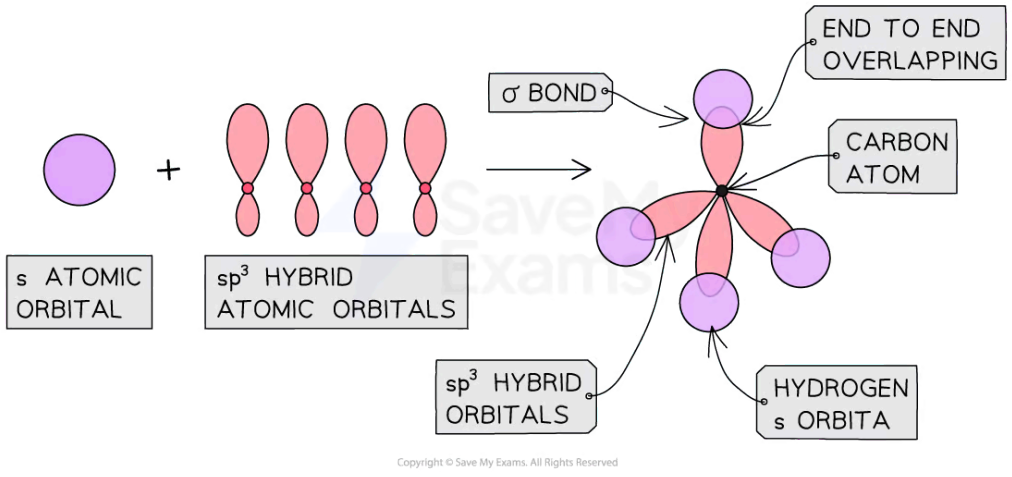
Orbital overlapping in bonds

- A single **covalent bond** is formed when two nonmetals share a pair of electrons
- Each atom contributes with one unpaired electron from a valence atomic orbital
- The overlapping between atomic orbitals form a molecular orbital
  - The overlapping can occur between **hybrid orbitals** too
- The type of the covalent bond depends on the type of overlapping
- There are two types of interaction:
  - **End to end overlapping** which form **sigma** bonds (**σ**)
  - **Sideways or lateral overlapping** which form **pi** bonds (**π**)
- Since a **σ** bond is formed by an end to end overlapping the bond is stronger and have greater bond energy than a **π** bond

Sigma bonds

- A **sigma bond (σ)** is formed when there is an end to end overlapping of atomic orbitals
- The interaction can occur between any type of atomic orbital (**non hybridized or hybridized**)
- The atoms are free to rotate around a sigma bond
- A single covalent bond is always a sigma bond
- g. In the case of methane (CH_4), the **four sp^3 hybrid orbitals** overlap with the **s-orbital** of the hydrogen atoms to form **four σ covalent bonds**

Sigma and Pi Bonds



Sigma bonds in methane

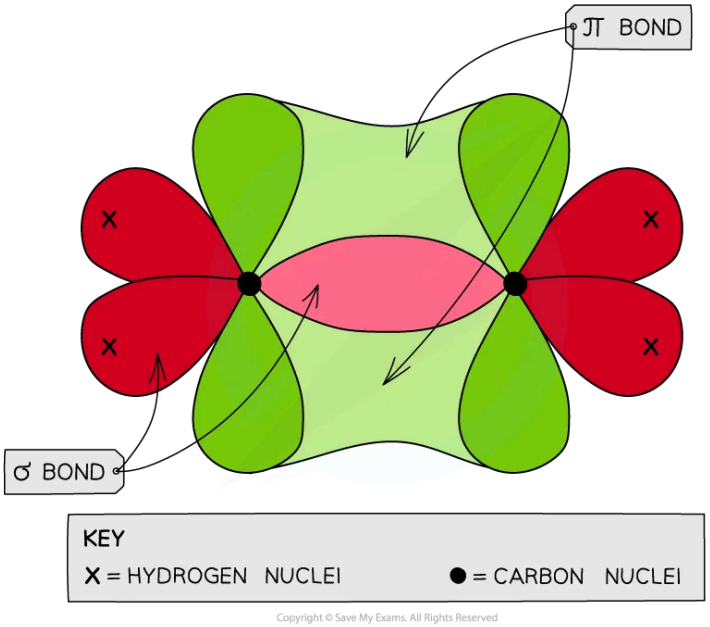
Your notes

Pi bonds

- A **pi bond (π)** is formed when there is sideways (lateral) overlapping of adjacent atomic p-orbitals
- The pi bond occurs perpendicular to the plane in which the sigma bonds are being formed
- A single **π** bond is drawn as two electron clouds, one above the plane and one below the plane of the sigma bond
  - The electron clouds above and below the plane prevents the rotation around the sigma bond
- **Double bonds** has always **one π bond** and triple bonds has always **two π bonds**
- g. In the case of ethene (C_2H_4),
  - Ethene has a double bond
  - The **two sp^2 hybrid orbitals** overlap end to end with the **s-orbital** of the hydrogen atoms to form **two σ covalent bonds.**
  - The **sp^2 hybrid orbitals** of both adjacent carbons overlap end to end to form **one σ covalent bond.**

While at the same time, the p orbitals of both adjacent carbon atoms sideways overlap to form **one π bond**

Bonding in Ethene



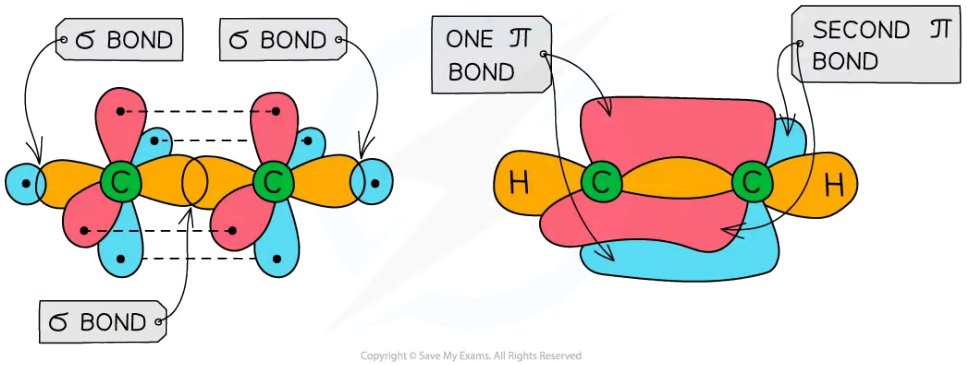
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Each carbon atom in ethene forms two sigma (σ) bonds with hydrogen atoms and one sigma (σ) bond with another carbon atom. The fourth electron is used to form a pi (π) bond

between the two carbon atoms

- E.g. In the case of ethyne (C_2H_2),
  - Ethyne has a triple covalent bond
  - **One sp hybrid orbital** overlaps end to end with the **s-orbital** of hydrogen to form **one σ covalent bonds.**
  - The **sp hybrid orbital** of both adjacent carbons overlap end to end to form **one σ covalent bond.**
  - While at the same time, the 4 p orbitals of both adjacent carbon atoms sideways overlap to form **two π bonds**

Bonding in Ethyne



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Ethyne has a triple bond formed from two π bonds and one σ bond between the two carbon atoms

Examiner Tips and Tricks

The number of pi bonds is always the number of bonds - 1. This occurs because there is always one sigma bond

Your notes